

Disruptive Biotechnology

Eric G. Meyer, PhD



Past and Future Promises of
Biomedical Engineering &
Wearables



PRESIDENTIAL COLLOQUIUM

APRIL 21, 2023

The Presidential Colloquium supports the teacher-scholar model of professional faculty activity which provides essential benefits in the education of our students, and creates and applies new knowledge to benefit society in general. In recognition that scholarship and teaching are mutually sustaining endeavors, the Presidential Colloquium promotes the scholarly engagement of Lawrence Tech faculty and the role it plays in enhancing our students' educational experience.

The Presidential Colloquium celebrates outstanding faculty and showcases their scholarship, research, and creative work in formal, invited lectures.

Tarek Sobh

President, Lawrence Technological University



Thank you President Sobh and Dean Grace for the invitation to speak and the introduction today. Thank you Matt Cole for organizing this event. Thank you to the audience for being here. This means a lot to me.

I have to start with my “dad joke”.

Why did the old man fall in a well?

- He couldn’t see that well!

I actually got that joke from ChatGPT.

My idea for this presentation was to ask: “What does the future of humanity look like from the perspective of Biomedical Engineering?”

– Disruptive Biotechnology

I wanted to start my story with the painting “The Garden of Earthly Delights” by Hieronymus Bosch. You can see in the outer panels it shows the Earth enclosed in a glass vessel. This painting is a triptych format which was common for the time period approximately 500 years ago. The painting was meant to be placed in a cathedral in the Netherlands.

* Edited transcript of the Presidential Colloquium seminar delivered at 10:30am on Research Day, the 21st of April 2023 in the Mary E. Marburger Science and Engineering Auditorium (S100), Lawrence Technological University

Full video: <https://youtu.be/ijkeznkovTE>



The inside panels depict the classic Christian story of Paradise on the left, the fall of man in the center and hell on the right. But what fascinated me about this painting (and caused me to write a 15-page term paper on this painting) was the alternate symbolism in these paintings. It is likely that the painter was an alchemist as he used symbolism and imagery from that medieval “science”. The primary goals of alchemists were to transmute base metals like lead into gold and to create the “elixir of life” to become immortal. While the first goal has been accomplished by modern physicists using particle accelerators, we are still working on the second.

My essay was for an Art History class that I took while studying abroad in Leuven, Belgium as an undergrad. It was a great class because we got to tour around Europe visiting historic cathedrals and museums.

That semester also gave me the motivation to go to Graduate school and led me to become a professor and stand before you today.



To all the students in the audience I want to tell you what a great experience it is to study abroad. If you get the chance, DO IT. It will change your life!

In the future, scientists may be able to give people enhanced abilities. Please indicate the extent you agree or disagree with people receiving the following changes that would:

John H. Evans
UC San Diego

	Strongly agree	Somewhat agree	Neither agree nor disagree	Somewhat disagree	Strongly disagree
Make people only need 4 hours of sleep per night	☐	☐	☐	☐	☐
Make people immune to infectious disease	☐	☐	☐	☐	☐
Make people have a healthy life for 140 years	☐	☐	☐	☐	☐
Give people much greater intelligence than humans have now	☐	☐	☐	☐	☐
Make people much happier	☐	☐	☐	☐	☐
Give people much greater physical strength than humans have now	☐	☐	☐	☐	☐

Before we discuss how to create the elixir of life, first we need to know if we should. Last week I was asked to participate in a Sociology survey from UC San Diego. I thought some of the questions were fascinating, so I wanted share them and to ask you to answer these for yourselves.

In the future, scientists may be able to give people enhanced abilities. Please indicate the extent you agree or disagree with people receiving the following changes that would:

Make people only need 4 hours of sleep per night.

Make people immune to infectious disease.

Make people have a healthy life for 140 years.

Give people much greater intelligence than humans have now.

Make people much happier.

Give people greater strength than humans have now.

So, please reflect on some of these. You don't need to provide or explain your responses. I think these are interesting social and ethical questions that Biomedical Engineers are engaged in.

My one criticism might be... I am not sure how much that "basic science" will provide to these solutions for humanity. I think there are other contributors and professions that make this topic interesting.

As I go through this talk, I want to share personal examples and thank important people in my life.

This is my grandfather Donald Meyer. He visited me when I was first starting out at as a professor at Lawrence Tech. I learned a lot about life and death from him. Not only did he instill my love of Ireland and St. Paddy's Day, but he lived 97 years and he even got to know my children. I didn't know any of my own great-grandparents, but I got to spend a lot of time with him when he was in assisted living and even while he took his last breaths.



He suffered from arthritis as most elderly people do. He had surgery to replace his knees with implants, which helped ease the pain and get him back moving again. He also had dentures, which he hated and said he wished he could have gotten dental implants instead.

These are some of my research topics, so I appreciated hearing from a patient who's life was improved with these advancements.

So jumping back to life extension myths... There are many versions of these in different cultures and throughout history. I think this is an important cross road between the humanities and STEM and the innovations we develop should be collaborative.





bostonglobe.com

Want to live to 150? The world needs more humans.

Believe it or not, we have an underpopulation problem to solve — and there's good reason to believe aging isn't inevitable.

By Raiany Romanni Updated March 29, 2023, 3:00 a.m.

This is a recent example of that social discussions in popular media. Do we really want people to live 150 years? I think most people would initial react negatively to this. Perhaps once they realize these would not be extra years of suffering from age-related diseases, but they could maintain their productivity and health span, they might be more supportive.

I know that I would have wanted my grandfather to have avoided suffering.

Another argument is that the Earth cannot sustain the human population if people live longer. But our birth rates have been declining for the past century, and the world is quickly moving towards negative population growth. So underpopulation could be a problem in the future.

These possible disruptions from a Biomedical perspective have historical precedence in other aspects of technology that we might be able to learn from.

Disruptive Technology



One of the examples related to Lawrence Tech and Metro Detroit is Henry Ford and the Model T. The assembly lines produced more than 15 million cars, which allowed regular people to replace their horse drawn transportation with the first mass market automobile.

In 1932 the Lawrence Brothers established our institution at the Ford factory in Highland Park to focus on technology innovations and to train the assembly line workers to be engineers.

Another example that I like to talk about is the 2007 unveiling of the first smartphone by Apple. There are now almost 7 billion similar devices in the world, allowing everyone the ability to connect and to access information.



My Background

Brandon HS / OSMTech

BS in Biomechanics - MSU

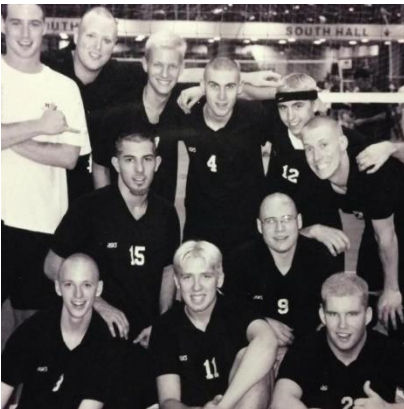
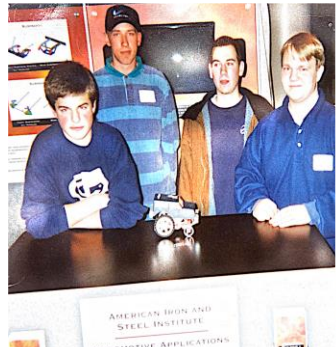
MS and PhD in ME - MSU

Post-doc Research Fellow - TCD



I will give you a bit of information about my background. I grew up in Oakland County (just like Lawrence Tech). I went to Ortonville Brandon High School, but I also attended a charter school for math, science and technology, called OSMTech.

Two of the teachers that had the biggest impact on my career to go into Biomedical Engineering were Mrs. Kyle Hughes and Mr. Mike Olsen. They were both founders of the school and they continue to teach at what is now called Clarkston Science Math and Technology.



After high school I went to Michigan State University. They didn't have a program in Biomedical Engineering I majored in Engineering Mechanics (which is kind of like applied physics) and also took premed courses and got a minor in biomechanics.

In addition to studying abroad, the other defining experience that I had at MSU was playing club volleyball.

When I came back from Belgium I was focused on getting into a Master's program and then ultimately got a PhD in Mechanical Engineering while doing applied research in the College of Osteopathic Medicine with topics of Orthopedics and Sports Medicine.



Then after my PhD degree my family and I moved to Ireland for two years, where I did a research fellowship at Trinity College Dublin.



This was focused on trying to solve the problem of treating arthritis using Tissue Engineering to regrow cartilage with stem cells.

I will briefly talk about all of these projects during my presentation.

Biotechnology



One of the biggest reasons that I wanted to go into Biomedical Engineering in the first place was motivated by science fiction and modern mythology stories around cyborgs and very advance prosthetic devices that work with the human body as if they are were a native limb.

As a lot of students think when they start out, my original plan was to get into prosthetics, which was what I thought Biomedical Engineers did. This is Marcus Rehm, who has the longest jump of any currently competing athlete. He is pretty unique even in the Paralympics he has a very advanced prosthetic device which is actually the leg that he jumps off of. That leg has been optimized, as a spring that stores and releases energy to give him that very long distance.

I had a summer internship working at Wright & Filippis. I thought I would be developing orthotics and prosthetics, but while I learned a lot about fabricating these devices it wasn't really Biomedical Engineering.

The other issue that I realized is, most of the patients getting these devices fairly low functionality. Actually, the biggest population of amputees is due to diabetes. This is a complication that results from poor blood flow and peripheral neuropathy with the later stages of that disease. These patients really don't have good outcomes, so I was kind of discouraged by that.

Biotechnology

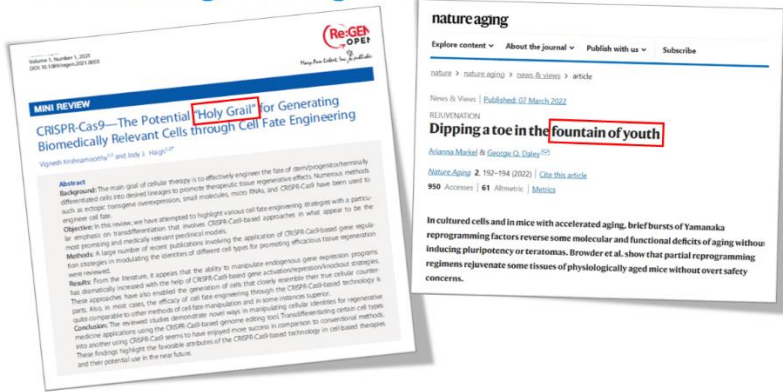


On the other hand, more recently I've become kind of fascinated by the advancing treatments for diabetes. This biotechnology development used recombinant DNA in the first genetic engineering product created by Genentech in 1978. Originally insulin was available for patients with diabetes from animal sources, but some patients developed allergic reactions to these forms. What they really needed was human insulin and this was accomplished by gene splicing into E Coli which then manufactured the protein that could then be isolated and refined. Another issue for these patients is controlling the amounts and timing of the insulin that they inject. Modern insulin pumps solve that problem by providing continuous insulin with customized dosages 24 hours a day.

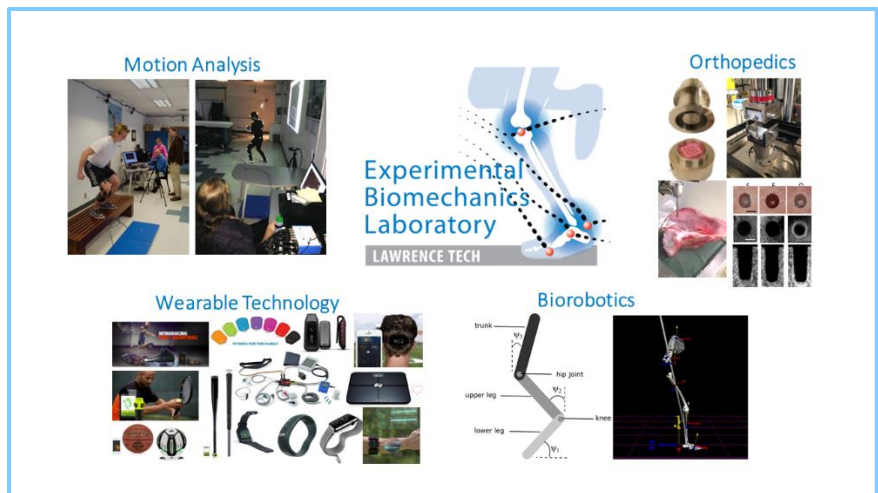
The other historical example of Biotechnology that I'm interested in is that of Brewing and other food production like for bread, cheeses and yogurt. Yeast are used to convert sugars or bacteria are used to break down proteins or fats. Humans have been using these technologies for 1000's of years and Stella Artois has been a brewery in Leuven Belgium that's been in operations since the year 1366! It's amazing to me as an American, how much history there is in other parts of the world.



Genetic Engineering



The tools available for genetic engineering have gotten even more powerful with CRISPR-Cas9 and other recently developed gene editing techniques. I wanted to share a couple interesting papers that even relate to my theme for this talk. I would also encourage you to read Walter Isaacson’s biography of Jennifer Doudna to learn more about how the Covid-19 vaccine developments were accelerated by these new technologies.



At Lawrence Tech I founded the Experimental Biomechanics Lab. This slide shows the topics that we are interested in and I will be sharing some

of these. In my talk today, I am not going to focus on the hard-core technical aspects of our research. If you are interested in that I would refer you to our website and my Google Scholar profile for a complete list of our more than 40 peer reviewed papers.



Eric G. Meyer
Director of Experimental Biomechanics Laboratory, [Lawrence Technological University](#)
Verified email at ltu.edu - [Homepage](#)
[Biomedical Engineering](#) [Biomechanics](#) [Mechanobiology](#) [Wearable Technology](#) [Sports Medicine](#)



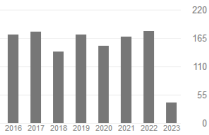
TITLE	CITED BY	YEAR
Anterior cruciate ligament injury induced by internal tibial torsion or tibiofemoral compression EG Meyer, RC Haut Journal of biomechanics 41 (16), 3377-3383	270	2008
Football playing surface and shoe design affect rotational traction MR Villwock, EG Meyer, JW Powell, AJ Fouty, RC Haut The American journal of sports medicine 37 (3), 518-525	198	2009
A Novel Osseous Densification Approach in Implant Osteotomy Preparation to Increase Biomechanical Primary Stability, Bone Mineral Density, and Bone-to-Implant Contact. S Huwais, EG Meyer International Journal of Oral & Maxillofacial Implants 32 (1)	188	2017
Excessive compression of the human tibio-femoral joint causes ACL rupture EG Meyer, RC Haut Journal of biomechanics 38 (11), 2311-2316	157	2005
Changing sagittal plane body position during single-leg landings influences the risk of non-contact anterior cruciate ligament injury Y Shimotochi, JP Ambegambikar, EG Meyer, SY Lee, SJ Shultz Knee Surgery, Sports Traumatology, Arthroscopy 21, 888-897	154	2013
Tibiofemoral contact pressures and osteochondral microtrauma during anterior cruciate ligament rupture due to excessive compressive loading and internal torque of the human knee EG Meyer, TG Baumer, JM Slade, WE Smith, RC Haut The American journal of sports medicine 36 (10), 1966-1977	132	2008
Syndesmosis fixation: analysis of shear stress via axial load on 3.5-mm and 4.5-mm quadricortical syndesmotic screws M Hansen, L Le, S Wertheimer, E Meyer, R Haut The Journal of foot and ankle surgery 45 (2), 65-69	125	2006
Low oxygen tension is a more potent promoter of chondrogenic differentiation than dynamic compression EG Meyer, CT Buckley, SD Thorpe, DJ Kelly Journal of biomechanics 43 (13), 2516-2523	124	2010

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Co-authors



Roger C. Haut
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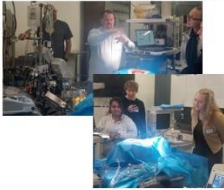


Mark R Villwock
University of Kansas



Daniel Kelly
Trinity Centre for Biomedical Eng...

<https://scholar.google.com/citations?user=3SbYkskAAAAJ>



I also want to tell the story of the Biomedical Engineering Department. We've been around for about 15 years and I think we have accomplished a lot in that time and provided a great service to our students and the University. I want to thank all of our alumni, the current students as well as my great colleagues. Our Chair Dr. Yawen Li, Dr. Michael Lancina, our awesome assistant Olivia Thompson, Professor John Peponis, and Dr. Hao Jiang.



Thank you for all of your work and support!

Arthritis Treatments

"Wait and See", RICE, NSAIDs

Injections (steroids, hyaluronic acid)

Cartilage repair

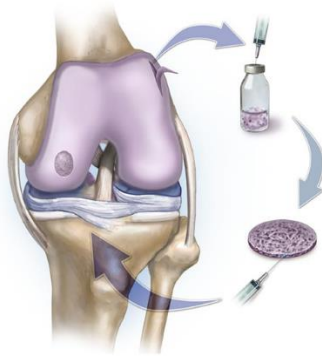
Resurfacing

Microfracture

Autologous chondrocyte implantation (ACI) or
"matrix assisted" (MA-ACI)

Total Joint Replacement

Stem Cells?



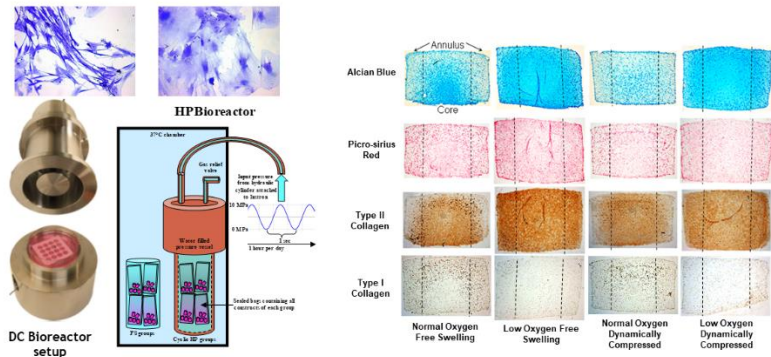
2 to 9 Year Outcome after Autologous Chondrocyte Transplantation of the Knee, Peterson et al. Clin Orthop Relat Res, 2000

Cartilage is one of the tissues that I have been most interested in, as it is quite troubling from a disease and treatment perspective. It doesn't have a blood supply or very many cells so it can't repair damage or regenerate once the process of arthritis degeneration gets started. The treatments available for cartilage are nowhere nearly as effective as those for ligament tears or bone fractures. One idea we had was to try to prevent cell death in the cartilage when an injury happens near the joint, like an Anterior Cruciate Ligament (ACL) tear. If you could save those cells within a day then they could have long term outcomes to preventing degeneration.



These are some of my colleagues at MSU in the Orthopedic Biomechanics Lab and I owe them so much as well!

Cartilage Tissue Engineering

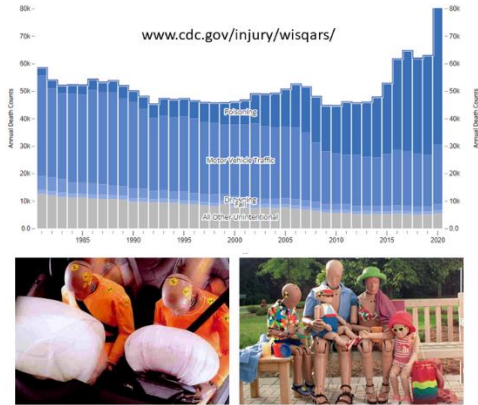


We used the Mesenchymal Stem Cells (MSCs) with a hydrogel that we formed into small cylindrical discs to make tissue engineered cartilage. Then we cultured those discs in a variety of environments for 6 weeks to see if we could get a functional tissue. The biochemical stimulation included steroids like dexamethasone and growth factors, like TGF- β 3. We also provided different types of mechanical loading, such as 10% compressive strain with an actuator built into our bioreactor or 10 MPa of hydrostatic pressure with a pressure vessel. These loads are typical for what your cartilage experiences every time you stand up or take a step. Finally, we also controlled the oxygen in our culture environment, and that actually turned out to be very important to produce cartilage with stiff and strong extracellular matrix materials like the glycosaminoglycans which are stained in blue and type-II collagen stained in orange.

I was lucky to get to go back to Ireland a couple years ago for the World Congress of Biomechanics, which was being hosted by my former post-doc adviser, Dr. Daniel Kelly. This is also Dr. Stephen Thorpe, who was a PhD student while I was in Dublin and is now a professor at the University College London.



Biomechanics

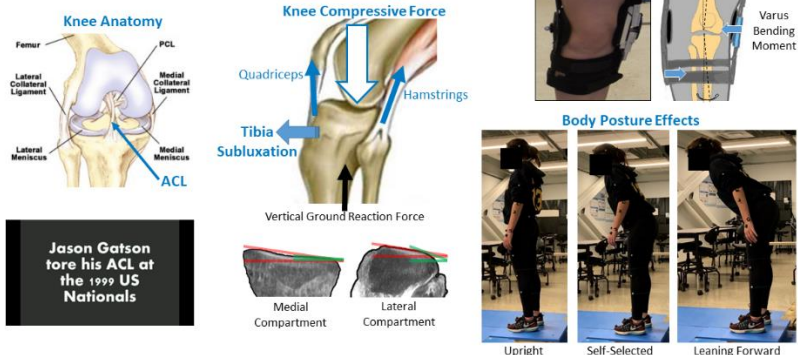


Switching gears to the story of impact biomechanics and our local contributions from the Motor City. Most of you are probably aware of the safety systems that have been developed for modern vehicles. These are things like seat belts, airbags, crumple zones and a reinforced occupant cabin. In order to design and test these systems, engineers needed a way to measure the forces on drivers and passengers and understand how injuries happened. They couldn't do these experiments with living people, so they made crash test dummies that have the correct sizes and behave like a human body during a crash. They are also instrumented to measure the impact loads and effectiveness of the new safety systems. These efforts have made modern cars much safer (at least until recently when distracted driving has started to reverse some of these fatality statistical trends).

Our students in the Biomedical Engineering Society took a tour, a couple weeks ago, of a local company called Humanetics which supplies almost all of the crash test dummies that are used world-wide. We got to hear from 3 of our BME alumni that work there as engineers.



Sports Injury Prevention



My PhD dissertation started out focusing on impact biomechanics, but I was interested in injuries in sports instead of car crashes. In this video you can see an example of Jason Gatson who tore his ACL twice during jump landings (the second time he was even wearing an injury prevention brace). Fortunately, this story has a happy ending as Jason went on to compete at the 2004 Olympics in Athens on his reconstructed ACL and didn't suffer another injury.

I used impact biomechanics methods to study ACL tears and high ankle sprains. In sports, there are frequently large forces in the lower extremity during noncontact situations when the player is in a compromising position. During a jump landing the ground reaction force is many times the player's body weight. In order to stay upright the knee has large muscle forces in the same direction as the ground reaction force. Newton's 3rd law says that in order for the system to maintain static equilibrium there must be a reaction force that is equal and opposite. So, the knee joint has huge compressive loads. But, the tibia is not flat and this creates an anterior shear force that loads the ACL.

At Lawrence Tech I wanted to try to prevent these types of injuries. One approach was to have athletes use a knee brace, but as you saw with Jason Gatson, that doesn't always work. Most injury prevention braces restrict extension and valgus bending rather than compression loading. So, we designed a brace that would transfer knee compression through

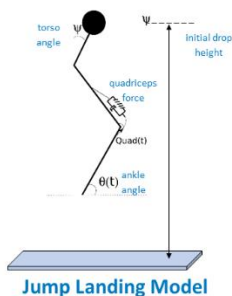
the medial compartment (inside) rather than the lateral side, which is less stable. But athletes don't like wearing these braces. They are uncomfortable and may affect their performance.

Another option is injury prevention training. For this we need to know the biomechanical best ways to train athletes.

We have also worked with the LTU athletics department and athletic trainers to assess players so that they can track and customize their injury prevention training.



Applied Biomechanics



Recent research in the lab has studied how the position of the upper body affects the knee reaction forces during jump landings. When you lean forward you have a softer landing, but if you are leaning backwards and especially if you lean to your outside when landing on one leg, then you can have large compressive forces in the lateral compartment which may exceed the injury threshold for the ACL.

I worked with Dr. Hamid Vejdani on this dynamic computational model to predict the knee reaction forces for a variety of different body positions. These thresholds could be provided in real time to warn athletes if they were in a risky position.

I am very proud to be a co-author for a number of posters that will be on display from 2:30-4:30 in the Atrium. This is a research project by Brian, who is one of our BME Master's students. He has been using the motion capture system to measure human movements to predict ACL risk.



Lawrence Tech
Biomedical Engineering

ACL injury risk factors of drop landing due to body position and athletic ability

Brian Czarnecki MS, Eric G. Meyer PhD

Biomedical Engineering, Lawrence Technological University, Southfield, MI



Experimental Biomechanics Laboratory

Background

- An estimated 250,000 Anterior Cruciate Ligament (ACL) injuries occur annually in the United States.
- Injury mechanism may be compression, valgus bending, or internal torsion.
- More than 70% do not involve direct player-to-player contact.
- Injuries may be caused by sudden deceleration, such as during landing sequence or cutting/pivoting.



Previous research examined the effects of different sagittal plane body positions during single-leg landings on biomechanics and muscle activation associated with risk for anterior cruciate ligament (ACL) injury. Landing styles included self selected, leaning forward, and upright landing.

Hypotheses

- Sagittal body position during jump landing will influence ACL injury risk. Higher GRF and knee loading during upright trials.
- Greater Athletic ability (Higher jumping ability) will lead to higher injury risk when leaning back, but much lower GRF values when natural landing or leaning forward landing due to better form. High risk/reward.
- One leg and two leg trials will show statistically significant variance. Standard deviation and errors for one leg trials will be greater.

	One Leg Standard Height	Two Leg Standard Height	Two Leg Variable Height
Jump Height	0.5m	0.5m	0.5m
Initial Angle	90°	90°	90°
Final Angle	90°	90°	90°
Initial Velocity	0 m/s	0 m/s	0 m/s
Final Velocity	0 m/s	0 m/s	0 m/s

Objective

The objectives in this experiment are to answer the following questions:

- Current drop landing research is performed at standard prescribed height (30cm, 45cm) with the only variance being sex, showing that Athletic ability is not taken into account. *Therefore, does Athletic ability also influence risk?*
- Subjects have a hard time maintaining balance/control during single leg drop landing trials, which can skew data. Subjects also may not be comfortable with single leg drop landings and therefore not perform naturally. *Therefore, do single leg or double leg trials differ?*

To answer these questions, the experiment will build upon previous research of sagittal body position influence by recreating original experiment AND:

- Varied drop heights that correspond to participants' athletic ability
- Drop landings with one leg or two leg trials

Materials and Methods

Vicon motion capture procedure: attach markers, calibrate system, T-pose, capture dynamic trials, analyze using Plug-In-Gait model tools, export .csv file for data analysis of joint kinematics.

Input Data:

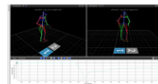
- Camera Position
- Marker Position
- Marker Color
- Marker Size
- Marker Shape
- Marker Material

Output Data:

- Joint Angles
- Ground Reaction Force
- Joint Torque
- Joint Power
- Joint Velocity
- Joint Acceleration

KEY Data points collected:

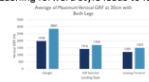
- Peak Ground Reaction Force
- Time to peak GRF
- Initial knee flexion angle
- Peak knee flexion angle
- Trunk angle
- Valgus bend angle



Results and Discussion

Data parallels previous research results:

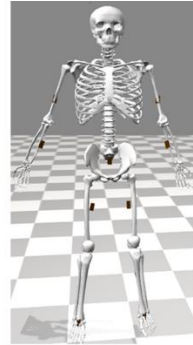
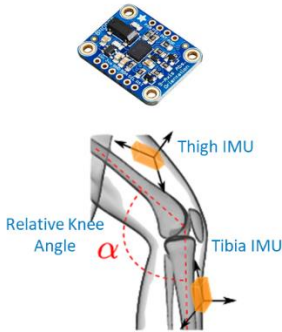
- Upright landing style leads to highest GRF (around 3X-4X body weight)
- Leaning forward style leads to lower GRF





Injury risk assessment is future goal, leveraging multiple data input sources to give real-time warnings and train athletes to prevent ACL and other lower extremity injury.

Inertial Sensors (IMUs)



This is Anthony Wash, who is a certified athletic trainer in addition to being a BME Master's student. He works full time doing functional movement screenings on athletes to assess their performance and injury risk and then he develops customized training programs based on that information. But we can't always bring athletes into the motion capture lab to do these assessments because it is time consuming and most trainers do not have access to a research lab. So they typically use qualitative scoring of the errors that are made during these functional movements.

Anthony will be presenting his research comparing these error scores with quantification of the joint kinematics that we can get from wearable sensors. These sensors are the same as what's in your Fitbit, but what we want is to place sensors on all the segments of the body to record all the motions of an athlete. The benefit of these sensors is they are small and light so they can be worn in the real world and many professional athletes are even wearing them during all of the practices and games throughout the season.

Our sensors also have Bluetooth so they can stream data directly to a smartphone to give real time feedback or to track changes in motion over time.



Experimental Biomechanics Laboratory



WEARABLE TECHNOLOGY INNOVATION CENTER

Integrating Wearable IMU Sensors to Collect and Analyze Human Movement Outside the Lab

Anthony Wash MS, Eric G. Meyer PhD

Biomedical Engineering, Lawrence Technological University, Southfield, MI



Lawrence Tech
Biomedical Engineering

Background

- There are many approaches when it comes to analyzing human movement. Quantitative examples include: optical motion capture and Inertial Measuring Unit (IMU) sensors, while the Functional Movement Screen (FMS) is a qualitative clinical technique.



Lab IMU sensor orientation

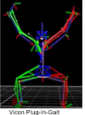


Wearable IMU sensors

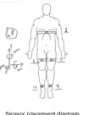


7 movements of the Functional Movement Screen

- Optical motion capture is the current gold standard for biomechanical analysis. However, many researchers, coaches and trainers do not have the access, time, and/or expense to assess subjects in a motion capture laboratory (1). IMU sensors are a cheaper and possibly more convenient way to quantify human motions during real world activities (2).



3D Motion Capture model



Sensor placement diagram



Subject T-pose for sensor placement



Deep squat with IMU sensors

Objective

- The objective of this research is to integrate wireless wearable IMU sensors with OpenSim software (3) to collect and analyze human movement. This method will be validated compared to motion capture. Then, real world human testing will compare quantitative IMU kinematics with qualitative FMS assessment by human movement professionals.

References: (1) Dowling et al. Inertial Sensor Based Feedback Can Reduce Key Risk Metrics for Anterior Cruciate Ligament Injury During Jump Landing, 2012. (2) Wu et al. Development of an Automatic Functional Movement Screening System with IMU Sensors, 2021. (3) Delp et al. OpenSim: Open-source Software to Create and Analyze Dynamic Simulations of Movement, 2007. (4) Shumway-Cook et al. Clinically Applicable Agreement Between the Xsens Motion Capture System and the VICON Motion Capture When Measuring Lumbar Region Inclination Motion in the Sagittal and Coronal Planes (2017)

Materials and Methods

- Standard Vicon procedure:** attach markers, calibrate system, T-pose, capture dynamic trials, analyze using Plug-in-Gait model tools, export .csv file for data analysis of joint kinematics.
- New IMU procedure:** attach sensors, start recording, T-pose and FMS movements, stop recording, data downloaded to smartphone and processed with python code, obtain graphs of joint kinematics.

Results

- The kinematic parameters of interest include hip ab/adduction, hip flexion, knee flexion, knee valgus and trunk relative angles (Figure 1).
- Kinematic results had moderate predictive value after regression analysis depending on the FMS activity (Table 1). In Deep Squat, participants who scored a 3 had greater HF and KF values for all movements compared to lower scores.
- Kinematic values varied slightly between left and right sides, so the minimum ranges of motion were used.
- Maximum values of hip abduction and knee valgus were used to compare those FMS scores because in order to score a 3, participant has to have less than 15° of hip abduction and knee valgus [2].
- Kinematic angles for Hurdle Step and Lunge were also compared with FMS scores. Results showed similarities between IMU and marker based kinematics.

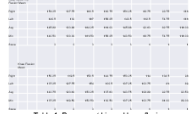


Table 1. Deep squat hip and knee flexion

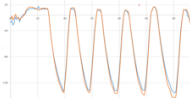
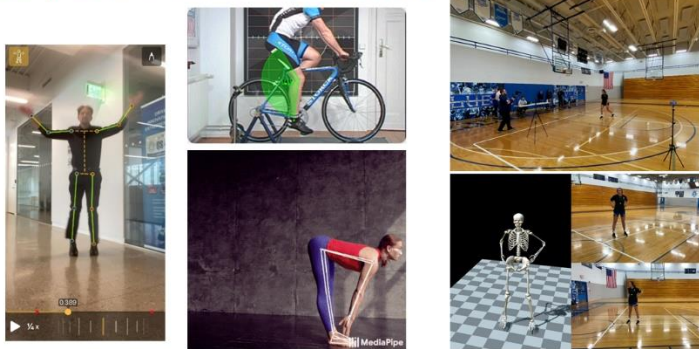


Figure 1. Representative deep squat hip flexion

Discussion

- IMU sensors do not require "line of sight" so FMS movements that are performed on the ground can be more easily tracked. They are also portable, to track motions in the clinic or during practices and games. Future research should attempt to collect data using a bigger sample size with diverse populations. Other movements, such as activities of daily living (ADL) and highly dynamic movements such as sprinting, jumping and cutting could also be studied (4). Sensor drift can be an issue with IMU data, and GPS or other types of sensors would also be necessary to track the position.

Computer Vision Pose Detection



Another approach to measuring human movement is with computer vision. Everybody has a very powerful camera that is built into your smartphone. It's actually capable of recording high definition and high-speed videos. So how can we use that technology to do motion capture?

A new system called OpenCap was developed at Stanford University and released as open source software last fall. You download an app on your phone that syncs between 2 and 4 iPhones to measure the body segment positions in 3D. Earlier this week we used OpenCap to help the LTU Women's volleyball team with some player assessments.

Nicole is another masters student who has been designing this system for various uses such as with coaches and trainers or even for STEM outreach programs to teach biomechanics to K-12 students.



Evaluating OpenCap for Collecting and Analyzing Human Movement with Smartphones

Nicole Carlin MS, Eric G. Meyer PhD
Biomedical Engineering, Lawrence Technological University, Southfield, MI

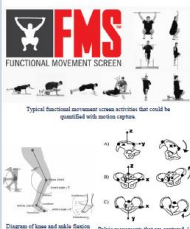


BACKGROUND

- Current methods for motion capture are costly, time consuming and advanced experience is required to collect and analyze the biomechanical motion data.
- OpenCap was developed as an open-source software for collecting 3D human movements using videos captured from smartphones.
- The web application allows users to sync videos from 2 to 4 iPhone cameras and uses computer vision-based pose detection to output the 3D movement of the human body.
- OpenCap may allow researchers, physical trainers, coaches and physical therapists to recruit their clients quickly and share the video analysis to get a visual feedback on what they are doing incorrectly.
- Educators can also use OpenCap in their classes to help students understand biomechanics of human movements.

OBJECTIVES

The goal of this project is to explore how OpenCap can be used for evaluating human movement in sports, education, aging or ergonomics. Comparisons were made with the current standard equipment used in biomechanics labs (Vicon Motion Capture). Participants simulated movements that would be typical done with a physical therapist or athletic trainer, including: Functional Movement Screen (FMS), lowering, walking, running, jumping and cutting. 3D motion of the body and the kinematics of the joints were calculated from both methods.



Typical functional movement screen activities that could be quantified with motion capture

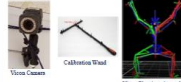
Diagrams of knee and ankle flexion and extension

Pelvic movement that are captured (A) from top (B) from side (C) from front

MATERIALS & METHODS

Vicon Motion Capture Procedure:

- The setup uses 10 cameras, 5-5 on each side to corner the capture volume area from all the sides giving a 3D view.
- Home software was used to collect all the data from the cameras and the force plate.
- Cameras were calibrated with a five marker wand that has markers on it.
- Vicon Plug-in-part biomechanical model is used for data analysis of markers on the subject to 3D joint kinematics.



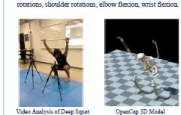
OpenCap Procedure:

- At least 2 iOS devices (2018 or later) were assumed as inputs with phone holders and their cameras pointed at the capture volume.
- A printed checkboard from OpenCap's web page was used for the initial capture volume calibration.
- Recorded videos are uploaded to OpenCap's server for processing using video pose detection algorithms. The processed motion data include the subject's 3D position and joint kinematics.



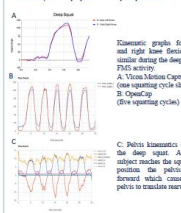
RESULTS

- Kinematic parameters were captured from both systems including pelvic translations and rotations, hip flexion, hip adduction, hip rotation, knee flexion, ankle flexion, ankle inversion, trunk rotation, shoulder rotation, elbow flexion, wrist flexion.



Video Analysis of Deep Squat

OpenCap 3D Model



Kinematic graphs for left and right knee flexion are similar during the deep squat FMS activity

A. Vicon Motion Capture
B. OpenCap (five repeating cycles)
C. Pelvis kinematics during the deep squat. As the subject reaches the up-leg position the pelvis tilt forward which causes the pelvis to translate forward.



Video Analysis of Picking

OpenCap 3D Model

During baseball picking the OpenCap 3D model frame rate was not fast enough to accurately track the subject's motion.

CONCLUSION

OpenCap demonstrated that it can gather the same data as the motion capture system. There are certain things that the system is able to do:

PROS	CONS
- Inexpensive - Short setup time - Portable - Video analysis	Video frame rate during fast paced movement

Who can benefit from it?

- Physical Therapist
- Physical Trainers
- Educators

FUTURE WORK

Things to Consider

- Frame rate
- Number of cameras
- Size of capture volume
- Activity being performed
- Data analysis on web application
- Work with other devices: devices iOS

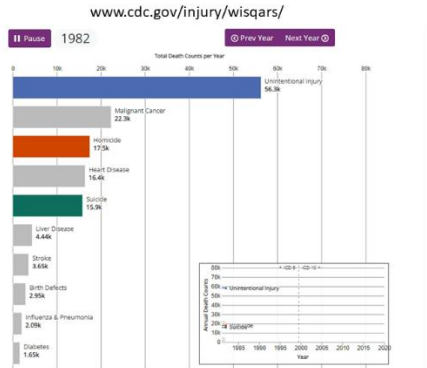
REFERENCES

- Chaffin et al. (2012) OpenCap: 3D Human Movement Dynamics from smartphone videos. ACOV program.
- Vicon Motion Capture Systems <https://www.vicon.com>
- OpenCap <https://www.opencap.ai>
- Kim et al. (2019) Kinematics on Ankle and Knee Joint of Post-Stroke Elderly Patients by Wearing Smart Elastic Band-Type Ankle-Foot Orthosis in Gait. Clinical Interventions in Aging

ACKNOWLEDGEMENTS

We are grateful to Department of Biomedical Engineering at Lawrence Technological University for allowing us to use its laboratory space and equipment to carry out this experiment.

Opportunities

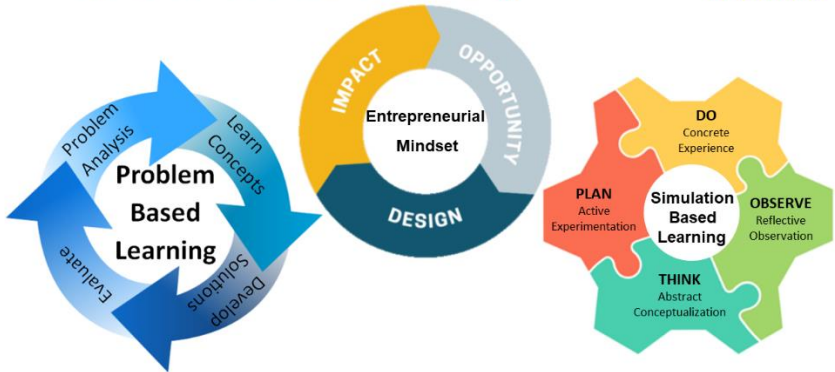


In addition to sports injury prevention, what are some other medical challenges that Biomedical Engineers can work on? There are still many diseases and injuries that are important in our society. We start our design process with opportunity recognition steps that are part of what has been termed an “Entrepreneurial Mindset”. The mindset and engineering skillset can be developed in students and we’ve implemented many curricular improvements at Lawrence Tech through the support of the Kern Entrepreneurial Engineering Network (KEEN for short).

Lawrence Tech has been one of the longest and most active KEEN partners and we had a very strong showing of faculty who attended the Annual Conference in Atlanta earlier this year.



Active Collaborative Learning



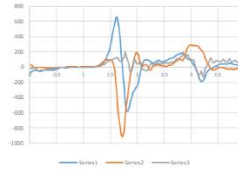
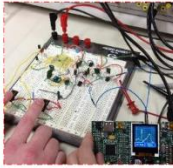
I also want to thank Dr. Andy Gerhart and other faculty who have been leading faculty workshops as part of KEEN to train new engineering professors with not only the entrepreneurial mindset, but more broadly the best practice educational strategies like Active Collaborative Learning, which can include modules that use Problem Based Learning, or Project Based Learning or Product Based Learning.

I have recently become interested in simulation-based learning, which has been applied for a long time in high risk fields like pilot training, and is becoming very common in medical training such as in Lawrence Tech's new Health Science programs (Nursing, Physician's Assistant and Cardiovascular Perfusion).

Shout out to Dr. Doug Melton who is a Program Director at KEEN and has provide us with tremendous support over the years.



Learn by Doing



Our approach has been sort of a learn by doing approach. This first example is 3 methods for making a EKG monitor. On the left is a breadboard circuit, which is what Prof. Peponis uses in the bioinstrumentation lab to get students to learn how to build and analyze circuit designs. On the right is a commercial EKG monitor which is a FDA regulated medical device and has actually been built into many smartwatches. But the problem with learning from these devices is that it is really hard to see how they work and understand it from a science or engineering perspective.

Another problem with both of those is it's very difficult to make changes if you wanted to modify the EKG system for a student design project. In the center is an "Open Electronics" kit that uses an Arduino microcontroller with a plug and play EKG sensor that can be programmed with computer code to do specific applications. Students can more easily use these resources in their senior capstone design or other projects that they are working on.

We wanted to give our students more experiences with these kinds of open electronics and programming to get more comfortable with them.



The LTU Softball coach has contacted you because of your expertise in biomechanics.



This is an example of a gyrometer that we used to build a swing sensor like for baseball, tennis or golf. We used the entrepreneurial mindset to motivate this design project in my biomechanics class. We even brought in the LTU Softball coach one year to give the BME students a hitting lesson so they would better understand swing kinematics. And we used the coach as the customer for this product, so the students would think about how to develop a useful solution for this opportunity.

We've also shared this with a few other partners in KEEN and published these strategies in ASEE and on KEEN's Engineering Unleashed platform.

Orthopedic Surgery Simulations



In our Orthopedics elective class we use similar entrepreneurial mindset modules and simulation based learning to introduce various surgery applications and the orthopedic implants and tools that surgeons use. The students start by building their own lower extremity anatomy from foam models of bones by adding all the soft tissues, muscles, skin and even blood vessels and nerves around that. So rather than doing anatomical dissections of just memorizing the anatomy from a book, this is more of an engineering approach to learning anatomy by building it. Once we have the lower extremity built, then we break it, and then we fix it. That is also an important part of the engineering approach.

A lot of the tools are just off the shelf drills and saws and other materials from Home Depot. This is actually pretty similar to how Orthopedic Residents are trained as well. You wouldn't want a surgical resident to just start learning on a real patient. They start with these Sawbone models and then progress to cadavers or animal tissues, to get a lot of practice before they work on humans.

This was at our Spring Exploration Day, where we do these surgical simulations with high school students. They get to do an ACL reconstruction on a knee model. This year was quite fun because both my son Nolan (shown here) and my daughter Bella came to the BME session. I think they also enjoyed it!



Medical Devices

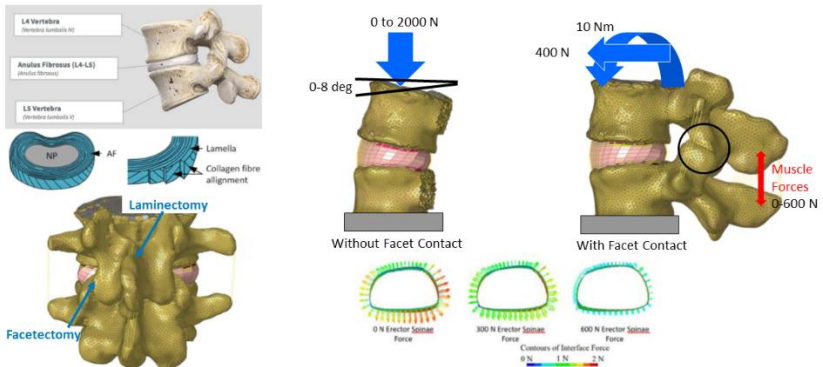


We've also partnered with the medical device industry. This is one example of a local company that makes spinal fixation implants and the surgical tools. When you have a damaged intervertebral disc (like a herniation) often that needs to be removed. You replace that with bone graft that allows the vertebral bones to fuse together as they heal. While the bones are fusing you need spacers and fixation from screws and rods attached to the back of the spine to hold them in place.

This system is great because these are real surgery tools that students can experience. We worked with the company to explore ways to make the surgery arthroscopic, so that it would be less invasive on the patients and to help them verify the FDA requirements for their devices by doing mechanical testing based on industry standards.



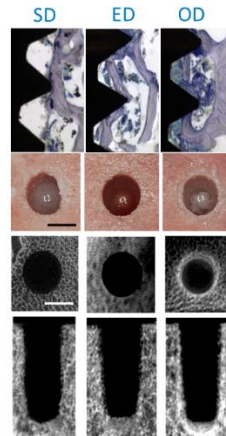
Biomechanical Modeling



Dr. Stephanie Rossman got her PhD from Lawrence Tech in Mechanical Engineering a little over a year ago. She built a finite element model of the lumbar spine and used it to understand how intervertebral disc herniations happen from bending and compressive loading. She also modified the model's anatomy similar to certain types of surgery like with a facetectomy or laminectomy when they have to remove parts of the vertebra or the muscle attachments. The disruption to the muscle forces actually caused significant increases to the loading in the intervertebral disc, which could lead to future problems for the patient after these types of surgery.



Implants



<https://versah.com/biomechanical-studies/>

Another research project that I worked on was with the dental implant industry. Versah was started by a Periodontist, Dr. Salah Huwais from Jackson, MI after he invented a new method for placing dental implants. In the past, dental implants were inserted by first drilling out the bone to create space. But in low density regions like in the posterior mandible, this reduces the implant stability as you are removing the bone tissue. His idea was to preserve and improve the mechanical properties of the bone by compacting it into the hole rather than extracting it with a drill. You can clearly see the dense layer of bone around the implant in these histology and x-ray images.

We did a validation study to test the implant stability of this new procedure versus traditional methods. We did this testing in CIMR here at LTU and published our results in the International Journal of Oral Maxiofacial Implants, one of the top journals in the field.



Since then, they have been growing like crazy and Dr. Huwais travels around the world teaching people how to use his new method. The Versah tools are in 1000s of dental offices and have been revolutionary for certain types of implant placement procedures.

Wearable Sensors



We've recently focused on using wearable technology like these various types of sensors that can be used for design project focused on creating consumer products. Many of these same sensors are used in medical devices to diagnose or treat diseases, but they are heavily regulated by the FDA which may slow the development or restrict solutions for people's general wellness. Sal is doing a masters design project in BME to use EMG sensors for physical therapy or for fitness training



WTIC
WEARABLE TECHNOLOGY INNOVATION CENTER

Electromyography Reusable Wearables for Targeted Muscle Rehabilitation and Growth

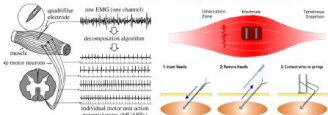
Salvatore Brauer, Eric G. Meyer PhD
Biomedical Engineering, Lawrence Technological University, Southfield, MI



Lawrence Tech
Biomedical Engineering

Background

- Electromyography sensors (EMG) measures small electrical signals generated by your muscles during contractions. These muscle activations produce movements, such as: lifting your arm, clenching your fist, or moving a finger. Nerve signals from the brain, produce a burst of electrical activity (which signal) that circulates through adjoining bone and tissue and can be measured by electrodes on the skin.

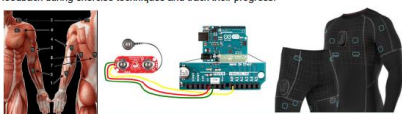


- EMG sensors are sensitive enough to pick up the slightest movements of muscles. EMG systems have been gaining popularity in recent years for use in both athletics and physical rehabilitation. By tracking the muscular activity along with motions, wearable technology systems can provide valuable clinical data that can be crucial in the process of rehabilitation or exercise.



Materials and Methods

- EMG sensor placement relative to the muscle anatomy is important for data collection.
- Different open electronics, Arduino-based EMG sensors and filtering techniques were evaluated to help with assisted rehabilitation.
- A prototype wearable system with embedded EMG sensors will be created for physical therapists or patients who need muscle rehabilitation.
- The sensor data will be compared to existing consumer products, such as the Athos wearable fitness tracker.
- The cost and usability of the wearable systems will be improved to give patients real time feedback during exercise techniques and track their progress.

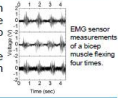


Objective

- The objective of the project is to use EMG sensors to target rehabilitation and create custom plans for people suffering from muscular degenerative diseases or for injury rehabilitation.
- A wearable device could help improve therapy and shorten recovery time by targeting and tracking specific muscles in the affected area.

Results

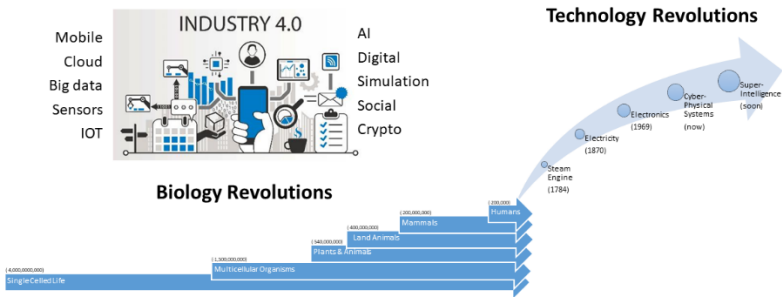
- Testing for this system is ongoing. Preliminary studies have shown promising results for general muscle movements. However, more sensors are needed (the Myo system has 8 EMG sensors), to monitor muscles firing patterns for targeted treatment.
- Advanced digital filtering techniques are necessary to eliminate external and internal noise and to analyze signal information throughout multiple muscles in the target area.



Discussion

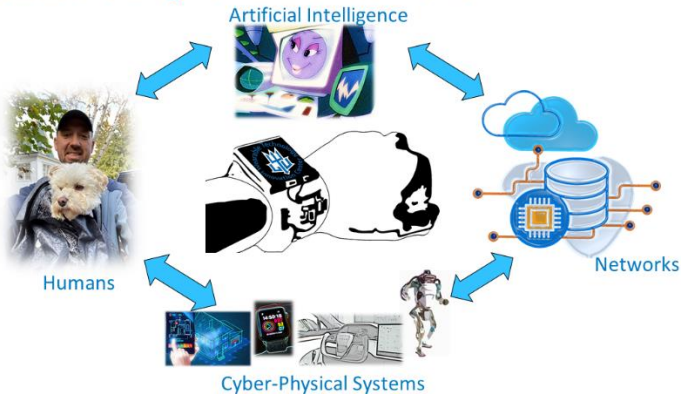
- Existing fitness suits with EMG sensors are extremely expensive and targeted for commercial use.
- They do not address some of the issues facing consumers, such as anatomy differences, maintaining sensor contact and reusability.
- Eventually, these wearable EMG sensors could be used by:
 - Fitness trainers
 - Rehabilitation
 - At home use

Disruptions



Back to my title theme of Disruptive Technology. There have been many Biological Revolutions driven by evolution and natural selection. Humans have not been around for very long. We have developed remarkable tools including fire, tools, time keeping, agriculture, writing. But it's really in the last few hundred years that industry has accelerated this innovation process. The first industrial revolution was textile mills and steam engines. Then we had electricity, motors and the assembly line in the early 1900. Recently we had electronics and computers. The current "Industry 4.0" revolution includes cyber-systems that are connected through mobile and cloud to remote sensing with big data and AI. There is also aspects of simulation and 3D fabrication through computer aided design.

Collaborating for Enhancement

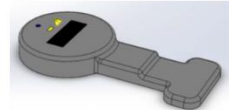
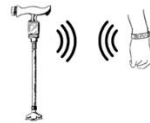


We've continued developing our plan with what is now called the Wearable Technology Innovation Center. We have submitted a NSF grant for creating a Regional Innovation Engine at Lawrence Tech. How can we merge the Industry 4.0 technologies into wearable systems that enhance human wellness and productivity?

Artificial Intelligence has been a huge topic of interest this year with many recent advancements that use generative AI, such as ChatGPT, GPT4, Dalle-2, and Midourney. I was fortunate to be part of a panel a couple weeks ago where we talked about our different approaches to incorporating AI into education at Lawrence Tech.



“Wearable” Senior Projects



We've also had senior projects that have incorporated wearable technology. These examples include, an app-based device for carpal tunnel therapy, a knee monitoring sleeve, a training device for CPR and wireless sensors for many applications

Many of these projects we've encouraged our students to present at National Conferences as well.



Industry Sponsored Projects



Other types of projects that I'm involved with are Contract Research projects with industry or student Industry Sponsored Projects. One example is a senior project that has been sponsored by Lear for the past 3 years. This year's team has made good progress toward their goals of creating a long-term comfort testing system for automotive seats and has a poster later today.



LEAR

Lucas Adams, Adam Kopp, Samantha Nott, Andrew Ulanak
Advisors: Eric G. Meyer PhD, Wu Pin-Zapadni PhD
Department of Biomedical Engineering, Lawrence Technological University

Lear Comfort System

Lawrence Tech
Biomedical Engineering

Project Goal

Design a new product: Comfort System designs to improve accuracy, reliability, and durability of the design while also facilitating rapid prototyping and testing for future improvement and iteration of the device.

Background

- Current assessment for long-term automotive seat comfort includes subjective human testing or non-representative simulation testing (Figure 1).
- Variables that are found to be most important in regards to seat comfort are pressure force, frictional force along the thigh, and vibration transmissibility.
- The existing prototype (Figure 2) incorporated some of these parameters, with vibration sensing, by use of a variable thickness foam mimicking large, pneumatic pressure sensors, and cylindrical piezoelectric stress sensors.



Figure 3: A 3D model of the device showing the sensor locations.

Device Design

Software and Coding:

- The original Arduino code was translated to Python.
- The code was updated to record data from, save the data to an on-board SD card, and directly export as a CSV file for spreadsheet use.
- The original equation for the sensors was incorrect. The force sensors were coded to use a linear curve when they should have been using a logarithmic curve. The stress sensors were using a linear equation instead of the proper fourth-order. Once corrected, the sensors were re-calibrated accordingly to applied to human mechanical calibration.
- Start and stop code was added for data collection, preventing the on-board microcontroller from corrupting when data collection was stopped during the recording of a data point.
- Acceleration code was added to accomplish the vibration analysis.

In-House Testing:

- An in-house huffman was fabricated using 3D printing and a STL model provided by Lear.
- The huffman was weighed down using four 25lb bags of lead (Figure 4) and was placed to be accompanied by two 25lb bags of rock salt.



Figure 4: A photograph of the device with four 25lb bags of lead placed on top of it.

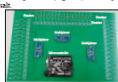


Figure 5: A photograph of the device with two 25lb bags of rock salt placed on top of it.

Prototyping

- The existing prototype was rebuilt to improve the durability of the design. 30 AWG silicon wire was replaced with 20 AWG TFE wire rated for automotive use. The polyurethane used was replaced with foam to create a more reliable and realistic design.
- Some-type stress sensors with spacers were replaced by plastic clamps to allow easy replacement of the sensors. These were also changed from cylindrical sensors to flat sensors. This change made it so that the stress sensors would not be affected by vertical loading to the same degree, collecting load parallel with the surface of the device instead. This also served to minimize issues with the cylindrical sensors whereas the sensors had a negative recovery time, a decrease effect, before returning to an appropriate zero point (Figure 6). The piezoelectric effect reduced the efficiency and accuracy of readings from the design.

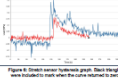


Figure 6: A graph showing the recovery time of the sensors. The y-axis is 'Voltage (mV)' and the x-axis is 'Time (s)'. The curve shows a sharp drop followed by a gradual recovery.

Sensor Calibration and Validation

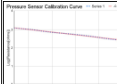


Figure 7: A graph showing the relationship between pressure and voltage. The y-axis is 'Pressure (kPa)' and the x-axis is 'Voltage (mV)'. The curve is a smooth, increasing line.

Figure 8: A graph showing the relationship between weight and voltage. The y-axis is 'Weight (N)' and the x-axis is 'Voltage (mV)'. The curve is a smooth, increasing line.

Figure 9: A graph showing the relationship between weight and voltage. The y-axis is 'Weight (N)' and the x-axis is 'Voltage (mV)'. The curve is a smooth, increasing line.

Limitations, Conclusions, and Future Work

Limitations:

- Due to the poor quality of the microphones, there was a significant amount of aliasing, so much so that sensor data was a mix of multiple sensor types. This resulting in data collection being limited to no data point every 45 seconds for a given sensor.
- The microcontroller internal storage was being pushed to the limit, which limits the possibility to add more components to the design.

Conclusions:

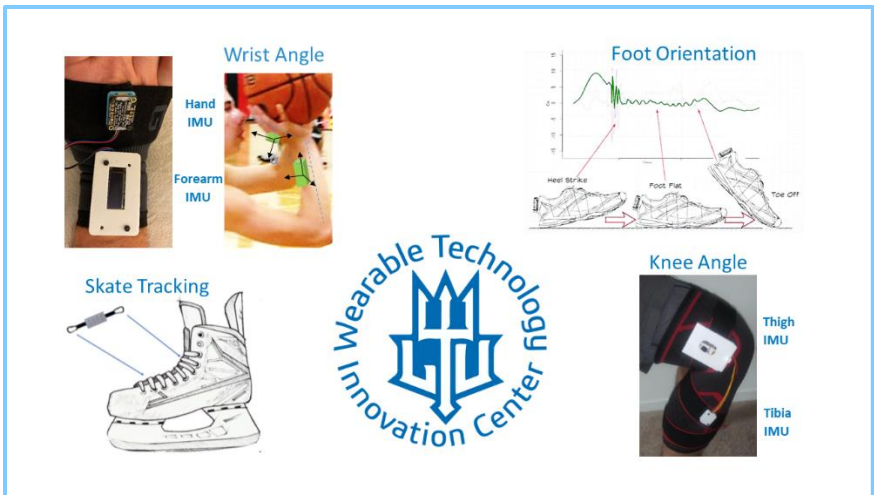
- The new code for the sensors corrected the incorrect equations to give correct data readings.
- The device material improved device durability and reliability.
- The stress sensors reduced noise in the readings and improved the recovery time.

Future Work:

- Replace the three microphones with a single 64-channel data acquisition system (DAQ).
- Develop a module for interfacing a computer or other device using the sensor readings.
- Modify the device's performance using human engineering.

Acknowledgments

We would like to thank our industry sponsor, Lear Corporation, for the opportunity to work on this project and help to improve our skills as future engineers. We would also like to thank our Lawrence Technological University and the Biomedical Engineering Department faculty for their efforts and support on this project. A special thanks to our industry advisor, Dr. Wu Pin-Zapadni, for his efforts in helping us complete this project.



We've also partnered with the Centropolis Accelerator and their start-up companies coming from outside the University or our students' own ideas. That's been exciting to work on these proof of concept devices with the ultimate goal to create commercial products.

Here is a swing sensor that also incorporates pressure monitoring in a tennis racket for monitoring tennis elbow. This poster will be presented by our BME master's student and LTU tennis athlete Carolin.

Forecasting Tennis Elbow Using Pressure Sensors

Carolin Minners MS; Eric G. Meyer PhD
Biomedical Engineering, Lawrence Technological University, Southfield, MI

Background

- Lateral epicondylitis, microscopic tears in the tendon connecting the bony part of the elbow to the muscle in the forearm, is diagnosed by about 50% of all tennis players.¹
- Mainly caused by repetitive arm movement
- Tennis elbow is reduces grip strength

Location of the bursa, tendons and inflammation

Tennis rackets based on angular velocity and acceleration direction

- Prevention techniques commonly include; a good warmup, gentle stretch before playing and using lightweight racquets with bigger grip sizes.²
- The gold standard for testing tennis elbow is to apply gentle pressure to examine your lateral epicondyle and the area above it. In some cases, it will be officially diagnosed using an MRI.
- Toppin, slice, or flat shot can be differentiated by angular velocity and acceleration directions.¹

Objective

- The objective of this design project is to integrate FSR force sensors and a microcontroller board with inertial sensors. Arduino software collects the pressure and motion data to analyze swing patterns, speed and grip pressure. The device will be tested by healthy and injured players to make predictions on the loss of grip pressure connected to lateral epicondylitis.

Previous Design and Function Ideas

Adafruit Feather board³

Preliminary functional diagram

Mechanical sketch for board and sensors placement in tennis racket

Development Prototype -> Final Version

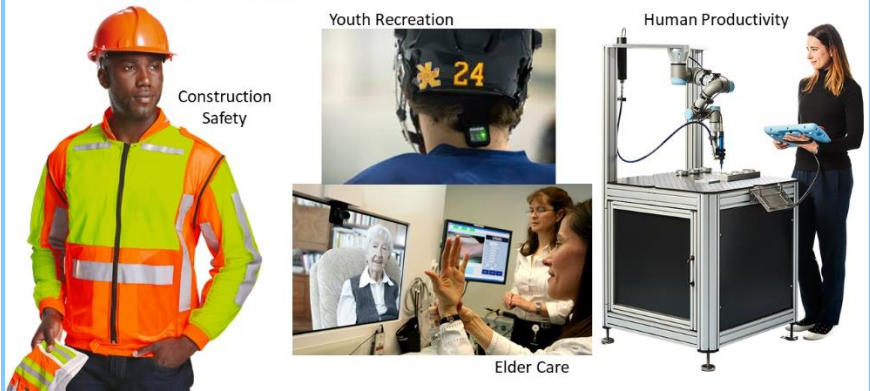
- Device functions:** attach sensors, turn the device on, sensor calibration of maximum force, start playing, shot detection with angular velocity threshold, measure racket speed, collect force data and compare it to calibration measurement, and red light when danger of injury.
- Adjustments prototype to the final design:** LED strip, smaller design to fit into the grip, solder wires to sensors and board.

Future Improvements

- Bluetooth connection and mobile app
- Smaller board size and weight
- Vibration capturing damper
- Detecting volleys and serves

References: 1) Shultz, Mayon, Medical Health Navigation: 2020
2) Skulnikova-Paschkevska et al. Sensor Technology for Sports Science: 2020 3) Adafruit, Adafruit: 2022

Societal Issues



The Wearable Technology Innovation Center has identified four social issues that we think can be addressed with wearable technologies. These include; construction and workplace safety, youth recreation and sports injuries, elder care and improved human productivity

Vaishnevi has built a haptic feedback system that could be incorporated into a wearable robotic assistant for pedestrians and workers. The ultimate goal is to help people navigate complicated environments and increase their awareness of dangerous objects. “Autonomous Walking”



Design and Testing of a Wearable Robotic Assistant

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Biomedical Engineering, Lawrence Technological University, Southfield, MI



Background

Workers in retail, construction sites, factories or warehouses may experience distractions that can lead to an accident or injury. Maintaining a high level of situation awareness can be difficult or reduce the worker's productivity and lapses may have severe or permanent consequences.

Tracking movements and helping people navigate safely could help people with disabilities such as blindness but could also improve workers productivity and reduce occupational risks.

Intuitive and helpful feedback, such as haptic mechanisms are needed to provide information on location and the surroundings to workers.



Objective

The project aims to develop a simple wearable robotic assistance device to help pedestrians navigate and avoid objects in workplace environments.

The goal is to not only increase the level of awareness for distracted walkers but also increase the detectability of potential threats that may not be noticed by attentive workers.

Specific aims of this proof of concept are:

1. Development of haptic feedback mechanisms.
2. Integration in a wearable device.
3. Demonstrate navigational assistance to a pedestrian.



Design and Prototypes

The prototype uses open computer hardware systems like Arduino with plug-and-play electronic vibrators for feedback, motors for direction and transceivers to navigate. The main focus for the proof of concept device are the haptic feedback mechanisms.



Feedback Mechanism

Controlling the feedback mechanism from a distance requires transceiver communication which can be built using this circuit.



Feedback Mechanism circuit



Remote control Mechanism



Remote control Mechanism Circuit

Applications and Future Work

User testing of the prototype showed that vibration and tension straps applied to the waist were effective haptic feedback mechanisms for steering and controlling the speed of a blindfolded pedestrian by a remote observer/controller.

The final proposed Wearable Robotic Assistant for Pedestrians (WRAP) system will adapt technologies developed for autonomous driving to human walking and provide feedback with the haptic actuators as demonstrated in this proof of concept device. Sensors (GPS, cameras, accelerometers, etc.) will monitor the user's environment, detect relevant information about objects and safety risks and communicate recommended actions to the user via an effective robot-human interaction framework. The system will augment the human body while performing simple, repetitive neuromuscular tasks like walking, stepping up or down, or picking up and setting down objects.

References

1. Haptic Feedback Evaluation Kit User Manual.
2. Lavi et al. IEEE International Symposium on Safety, Security, and Rescue Robotics, 2013.
3. Mwakalonge et al. J Traffic Transportation Engineering, 2015.
4. Liu et al. IEEE Int Conf Robotics Automation (ICRA), 2015.

Wearable Technology Innovation Center



So far, we have developed partnerships with over 30 local and regional companies, hospitals, non-profits, and government agencies as part of this NSF proposal. We will be continuing to develop collaborative resources to innovate and create new solutions, supporting industry and training workers to get into wearable technology design. Please see our webpage and reach out to us if you are interested in being a partner.

Lawrence Technological University



Wearable Technology Innovation Center (WTIC)

Lawrence Technological University (LTU) is establishing Michigan's first Wearable Technology Innovation Center (WTIC). Through the planning and establishment of WTIC, innovators will access resources provided to design, research, develop, manufacture and commercialize their technologies and products, as well as establish and scale their companies. These new consumer products will combine artificial intelligence, big data analytics, robotic sensing, mobile networking, tracking and navigation into assistive devices that augment the human body.

WTIC will encompass a robust entrepreneurial ecosystem, innovation center, and regional economic development engine. It will coalesce innovative companies, entrepreneurs, start-ups, manufacturing firms, industry leaders, university researchers, and workforce development stakeholders to foster the development and production of wearable technology solutions in the areas of:

- Elder Care
- Worker Safety and Protection
- Youth Recreation and Athletic Safety
- Human Productivity



WEARABLE TECHNOLOGY
INNOVATION CENTER
LAWRENCE TECHNOLOGICAL UNIVERSITY

Thank you! (not a complete list)

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I would like to thank my colleagues! This is a long list. I apologize if I left anyone off. This includes my colleagues from Lawrence Tech and also my colleagues from MSU and Trinity College.

I would also like to thank our students. We have many students in BME, not only in the Master's program, but also undergraduates and our alumni that have worked on research projects with us.

Glucose Monitoring



To conclude, I want to go back to the diabetes theme. Glucose monitoring has been around since the late 1960's, but that often involved finger sticks that needed to be done many times a day to adjust the insulin injections for patients.

More recently continuous glucose monitoring has been shown to be much more effective, but these systems started out as large devices that were used in hospitals. Recently that technology has been miniaturized into wearable devices that are part of the medical care for these patients. They connect over Bluetooth to a smartphone or even a smart watch.

I feel like this continuous glucose monitoring is really helpful for people like my wife's grandmother Virginia Marquardt who had diabetes as she got older (she also lived 97 years!). She was also very important to us in our lives. She introduced me, and my parents and now my children to the wonders of Petoskey, MI. Just north of Traverse City along Lake Michigan and we now spend a lot of time there as my parents have retired there.



Continuous Glucose Monitoring



INTERVIEWS
'It's been an education' - Rory McIlroy using skin sensor to monitor blood sugar levels



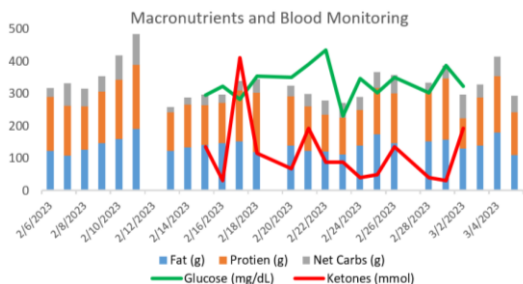
Wednesday, November 23, 2022



Continuous glucose monitors (CGMs) are becoming more of a consumer product. These were medical devices for diabetes patients, but now athletes have gotten interested in using these to understand their health and nutrition. Rory McIlroy wore a CGM at the PGA Championship last fall (which he won) as a way to improve his training and keep his body in peak performance throughout the season. You can sign up for app-based subscriptions that provide CGMs and the health and wellness support.

One problem with current CGMs is that they require injecting a small filament under the skin, which is what actually does the glucose sensing. But new technology may allow sensors that can monitor glucose or other biochemical noninvasively. And even Apple has been rumored to be interested in including this type of sensor in future Apple watches.

I have been tracking my own nutrition recently and even started doing some finger sticks to measure my blood glucose and ketone levels. I am not sure how helpful this information is to me yet. What I really need is a personal AI assistant that sorts through all the data and tells me when to eat and exercise and what supplements to take.



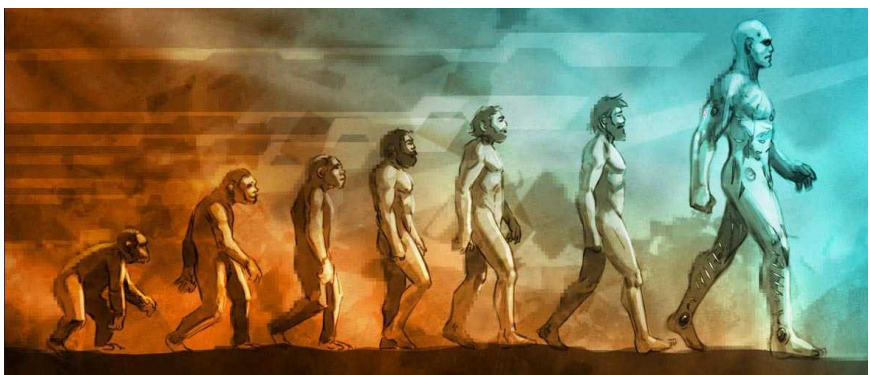
"if ever man fell ill, there was no defense—no healing food, no ointment, nor any drink—but for lack of medicine they wasted away, until I showed them how to mix soothing remedies with which they now ward off all their disorders."
-Prometheus Bound, Aeschylus 525-455 BC



To tie this all back to the original anti-aging theme. Here are a couple recent examples of using hormones or cytokines to treat or even reverse type 2 diabetes. GLP-1 is a potential target, that instead of just treating the disease with insulin may actually treat the underlying causes.

Another example is from Genentech (which created insulin and HGH) is working on using genetic engineering to reprogram cells to make them more youthful.

With that I will conclude that we really need to discuss the social and ethical issues of biotechnology and the applications that I have discussed here. But I do think this is a really exciting topic and I hope to live a long and healthy life.





I will leave you with a quick picture of Petoskey. I want to thank my parents Marcia and Kevin Meyer for instilling a love of learning. I also thank my children who are here. Bella, Nolan and Emmett hopefully will be Blue Devils very soon.

For my next study abroad, I have been dreaming about visiting Mars on the new SpaceX Starship. But I will want to come back and retire in Ireland or Petoskey with my wife Suzi. Even if I don't get to go to Mars, I appreciate that she was willing to pretend to be a Martian with me. Thank you, Suzi, for all of your love and support. None of this would have been possible without you!



Acknowledgements





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Eric G. Meyer, PhD is an Associate Professor in Biomedical Engineering and has directed the Experimental Biomechanics Lab (EBL) at Lawrence Technological University since 2010. Meyer also co-founded and directs the Wearable Technology Innovation Center (WTIC), a partnership between more than 20 LTU faculty and staff and more than 30 regional industry, academic, non-profit and government partners. The WTIC promotes Open Innovation for Human Enhancement and works with its partners and members of the community to coalesce and support a thriving innovation



ecosystem in Southeast Michigan. His research focus on preventing sports injuries also overlaps with engineering design projects that allow youth and recreational athletes to track their biomechanics during movement to customize their training using wearable sensors.

Meyer mentors engineering and premedical students to conduct translational research that combines in vivo testing of human and animal subjects with cadaver and surrogate in vitro models and computer simulations to understand normal health, acute injury and degenerative diseases. He enthusiastically cultivates collaborations with clinicians and the medical device industry to develop preventative and regenerative treatments for bone and soft tissues. Meyer has published one book chapter and 42 peer-reviewed articles that have received more than 2250 peer citations. Recently, he has presented and published a seminal biomechanical validation study of Osseodensification, a novel implant osteotomy preparation technique that increases implant stability with Dr. Salah Huwais, DDS, a practicing Periodontist, inventor and founder of Versah LLC. He has given more than 100 conference, workshop and invited presentations, including as a panel member for recent Humanities + Technology topics such as "AI and the Future of Education".

Meyer teaches undergraduate required courses; Introduction to Biomechanics and Biomechanics Lab, and Tissue Mechanics, as well as the elective Engineering Applications in Orthopedics and the graduate course Cellular and Molecular Mechanobiology. He helped the College of Engineering faculty committee to implement a cohesive, four-year multidisciplinary engineering design program focused on creating entrepreneurially minded engineers. Meyer has developed entrepreneurial-minded learning course modules and trained other faculty to use similar modules in their courses as part of grants from the Kern Family Foundation (KEEN). He has also developed Course Based Research Experiences for Undergraduates (CRE) projects as part of a LTU grant from the Howard Hughes Medical Institute. In addition, Dr. Meyer offers a number of outreach programs for high school students and advises many projects for undergraduate and graduate students.